Robotic Manipulation

Perception, Planning, and Control

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Note: These are working notes used for a course being taught at MIT. They will be updated throughout the Fall 2026 semester.

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Appendix A
Spatial Algebra

Throughout these notes we've introduced the basic rules of spatial algebra. I find myself looking back at them often! For ease of reference, I've compiled them again here. These make heavy use of our "monogram notation".

The Drake documentation also has a very nice summary of the multibody quantities and how they map to the monogram notation.

A.1 Position, Rotation, and Pose

As we introduced here, we use ${}^{B}p^{A}_{C}$ to denote a position of point or frame A relative to point or frame B expressed in frame C. We use ${}^{B}R^{A}$ to denote the orientation of frame A measured from frame B; unlike vectors, pure rotations do not have an additional "expressed in" frame. Similarly, we use ${}^{B}X^{A}$ to denote the pose/transform of frame A measured from frame B. We do not use the "expressed in" frame subscript for pose; we always want the pose expressed in the reference frame.

The basic rules of spatial algebra are as follows:

- Positions expressed in the same frame can be added when their reference and target symbols match:

$$ {}^{A}p^{B}_{F} + {}^{B}p^{C}_{F} = {}^{A}p^{C}_{F}. \tag{1}$$

  Addition is commutative, and the additive inverse is well defined:

$$ {}^{A}p_{F}^{B} = -{}^{B}p_{F}^{A}. \tag{2} $$

Those should be pretty intuitive; make sure you confirm them for yourself.

- Multiplication by a rotation can be used to change the "expressed in" frame:

$$ {}^{A}p_{G}^{B} = {}^{G}R^{F}\,{}^{A}p_{F}^{B}. \tag{3} $$

You might be surprised that a rotation alone is enough to change the expressed-in frame, but it's true. The position of the expressed-in frame does *not* affect the relative position between two points.

- Rotations can be multiplied when their reference and target symbols match:

$$ {}^{A}R^{B}\,{}^{B}R^{C} = {}^{A}R^{C}. \tag{4} $$

The inverse operation is also simply defined:

$$ \left[{}^{A}R^{B}\right]^{-1} = {}^{B}R^{A}. \tag{5} $$

When the rotation is represented as a rotation matrix, this is literally the matrix inverse, and since rotation matrices are orthonormal, we also have $R^{-1} = R^{T}$.

- Transforms bundle this up into a single, convenient notation when positions are measured from a frame (and the same frame they are expressed in):

$$ {}^{G}p^{A} = {}^{G}X^{F}\,{}^{F}p^{A} = {}^{G}p^{F} + {}^{F}p_{G}^{A} = {}^{G}p^{F} + {}^{G}R^{F}\,{}^{F}p^{A}. \tag{6} $$

- Transforms compose:

$$ {}^{A}X^{B}\,{}^{B}X^{C} = {}^{A}X^{C}, \tag{7} $$

and have an inverse

$$ \left[{}^{A}X^{B}\right]^{-1} = {}^{B}X^{A}. \tag{8} $$

Please note that for transforms, we generally do *not* have that X^{-1} is X^{T}, though it still has a simple form.

A.2 Spatial velocity

As we introduced here, we represent the rate of change in pose using a six-component vector for *spatial velocity*:

$$ {}^{A}V_{C}^{B} = \begin{bmatrix} {}^{A}\omega_{C}^{B} \\ {}^{A}\mathrm{v}_{C}^{B} \end{bmatrix}. \tag{9} $$

${}^{A}V_{C}^{B}$ is the spatial velocity (also known as a "twist") of frame B measured in frame A expressed in frame C, ${}^{A}\omega_{C}^{B} \in \mathbb{R}^3$ is the *angular velocity* (of frame B measured in A expressed in frame C), and ${}^{A}\mathrm{v}_{C}^{B} \in \mathbb{R}^3$ is the *translational velocity* (along with the same shorthands as for positions). Spatial velocities fit nicely into our spatial algebra:

- Velocities add when they are expressed in the same frame:

$$ {}^{A}\mathrm{v}_{F}^{B} + {}^{B}\mathrm{v}_{F}^{C} = {}^{A}\mathrm{v}_{F}^{C}, \qquad {}^{A}\omega_{F}^{B} + {}^{B}\omega_{F}^{C} = {}^{A}\omega_{F}^{C}, \tag{10} $$

  and have the additive inverse, ${}^{A}V_{F}^{C} = -{}^{C}V_{F}^{A},$.
- Rotations can be used to change between the "expressed-in" frames:

$$ {}^{A}\mathrm{v}_{G}^{B} = {}^{G}R^{F}\,{}^{A}\mathrm{v}_{F}^{B}, \qquad {}^{A}\omega_{G}^{B} = {}^{G}R^{F}\,{}^{A}\omega_{F}^{B}. \tag{11} $$

- Translational velocities compose across frames with:

$$ {}^{A}\mathrm{v}_{F}^{C} = {}^{A}\mathrm{v}_{F}^{B} + {}^{B}\mathrm{v}_{F}^{C} + {}^{A}\omega_{F}^{B} \times {}^{B}p_{F}^{C}. \tag{12} $$

- This reveals that additive inverse for translational velocities is not obtained by switching the reference and measured-in frames; it is slightly more complicated:

$$ -{}^{A}\mathrm{v}_{F}^{B} = {}^{B}\mathrm{v}_{F}^{A} + {}^{A}\omega_{F}^{B} \times {}^{B}p_{F}^{A}. \tag{13} $$

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A.3 Spatial force

As we introduced here, we define a six-component vector for *spatial force*, using the monogram notation:

$$ F_{\mathrm{name},C}^{B_p} = \begin{bmatrix} \tau_{\mathrm{name},C}^{B_p} \\ f_{\mathrm{name},C}^{B_p} \end{bmatrix} \quad \text{or, if you prefer} \quad \left[F_{\mathrm{name}}^{B_p}\right]_C = \begin{bmatrix} \left[\tau_{\mathrm{name}}^{B_p}\right]_C \\ \left[f_{\mathrm{name}}^{B_p}\right]_C \end{bmatrix}. \quad (1 $$

$F_{\mathrm{name},C}^{B_p}$ is the named spatial force applied to a point, or frame origin, B_p, expressed in frame C. The name is optional, and the expressed in frame, if unspecified, is the world frame. For forces in particular, it is recommended that we include the body, B, that the force is being applied to in the symbol for the point B_p, especially since we will often have equal and opposite forces.

Spatial forces fit neatly into our spatial algebra:

- Spatial forces add when they are applied to the same body in the same frame, e.g.:

$$F^{B_p}_{\text{total},C} = \sum_i F^{B_p}_{i,C}. \tag{15}$$

- Shifting a spatial force from one application point, B_p, to another point, B_q, uses the cross product:

$$f^{B_q}_C = f^{B_p}_C, \qquad \tau^{B_q}_C = \tau^{B_p}_C + {}^{B_q}p^{B_p}_C \times f^{B_p}_C. \tag{16}$$

- As with all spatial vectors, rotations can be used to change between the "expressed-in" frames:

$$f^{B_p}_D = {}^D R^C f^{B_p}_C, \qquad \tau^{B_p}_D = {}^D R^C \tau^{B_p}_C. \tag{17}$$

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